Mini Project
Kinematic calibration

Team members

Last name	First name	Last name	First name	
				Students, *third if any*

Prepared by:

Name	Signature	Date	
			Supervisor

Controlled by:

Name	Signature	Date	
			First team member

Approved by:

Name	Signature	Date	
			Second team member, *third if any*

Tests results confirmed by:

Name	Signature	Date	
			First team member

Tests results confirmed by:

Name	Signature	Date	
			Second team member *third if any*

Tests results approved by:

Name	Signature	Date	
			Supervisor

Degree Programme
Systems Engineering

Sion, April 20, 2026

Performance Qualification

1. Camera calibration.

PQ ID	Reset the system
1	From aborted state in execute mode after resetting the PLC program. One team member performs a **cold** reset and starts in a browser connected to CtrlX OS / /plc/status/applications: The state is aborted in execute mode yes / no? …

2. Camera calibration.

PQ ID	Validate calibration
2.1	The test engineer presses the sequence of hardware buttons: Reset – Reset - Start The position of the camera is on the QR code, (y/n)? …
2.2	An indicator prompts you to validate the calibration (y/n)?...
2.3	The test engineer stops the system and presses the sequence Reset – Reset – Start, there is no more prompts to validate the calibration (y/n)?...

3. Pick and place sequence

PQ ID	PackML command Stop
3.1	From aborted state, if calibration has been validated, the operators inserts a tube on nest position: (nest position)?... The operation presses the sequence Reset – Reset – Start, insert the nest position above in the User Interface and presses the **Start Pick** button. The robot picks the tube from position (position)?... The robot is in high position with the tube in the gripper (y/n)? …
3.2	There is a tube in the gripper, the operator selects the nest number (nest position)?.... The operator presses the **Start Place** button. The robot places the tube in the nest number (number)?.... The z axis is in up position (y/n)? There is a tube in the gripper (y/n)?

4. Fast commissioning

PQ ID	Check Hold
4,1	The supervisor opens the door, move the plate from **n** centimetres to the centre of the cell: (Number of centimetres)?....
4.2	The supervisor starts a chronometer. One team member modifies the PCS, enable or disable it with some buttons. Done (y/n)?... One team member reset the PLC using the sequence descried in PQ ID 1, Done (y/n)?... One team member performs a camera calibration validation as described in PQ ID, Done (y/n)?... Write the time of the chronometer here in minutes, seconds:….
4.3	The supervisor places a tube in the nest number (nest number) ?.... The supervisor starts a pick sequence from the nest number (nest number) ?.... The robot picks the tubs (y/n)?...